US012398778B2

(12) **United States Patent**
Li et al.

(10) **Patent No.: US 12,398,778 B2**
(45) **Date of Patent: Aug. 26, 2025**

(54) **SYSTEM FOR ABSORBING FLEXURAL WAVES ACTING UPON A STRUCTURE**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 941 days.

(21) Appl. No.: **17/536,582**

(22) Filed: **Nov. 29, 2021**

(65) **Prior Publication Data**

US 2023/0167877 A1 Jun. 1, 2023

(51) **Int. Cl.**
F16F 15/04 (2006.01)
F16F 7/10 (2006.01)

(52) **U.S. Cl.**
CPC ***F16F 15/04*** (2013.01); ***F16F 7/1005*** (2013.01)

(58) **Field of Classification Search**
CPC F16F 15/04; F16F 7/1005
See application file for complete search history.

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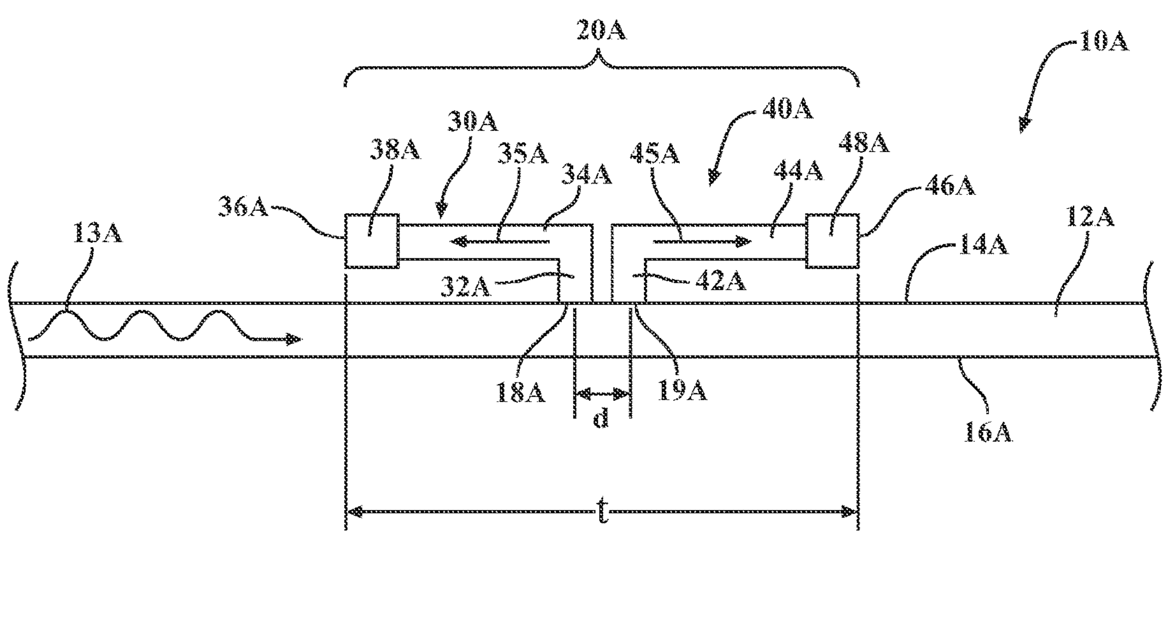
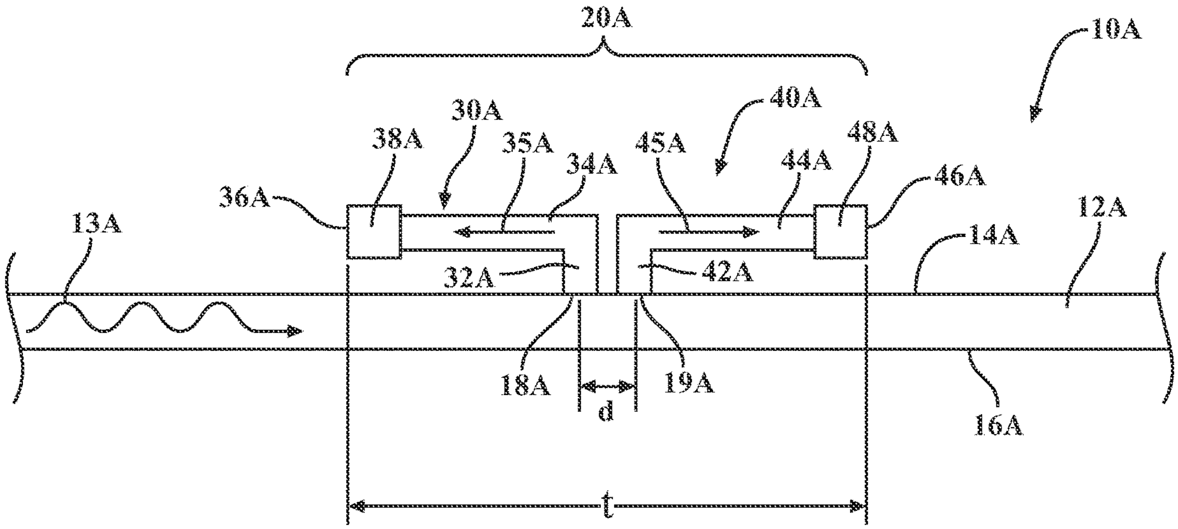
(57) **ABSTRACT**

Described are systems for absorbing flexural waves acting on a structure. In one example, the system includes a first resonator connected to a structure at a first location and a second resonator connected to the structure at a second location. The distance between the first location and the second location is based on a frequency of a flexural wave acting upon the structure and an orientation of the first resonator and the second resonator with respect to each other.

15 Claims, 6 Drawing Sheets

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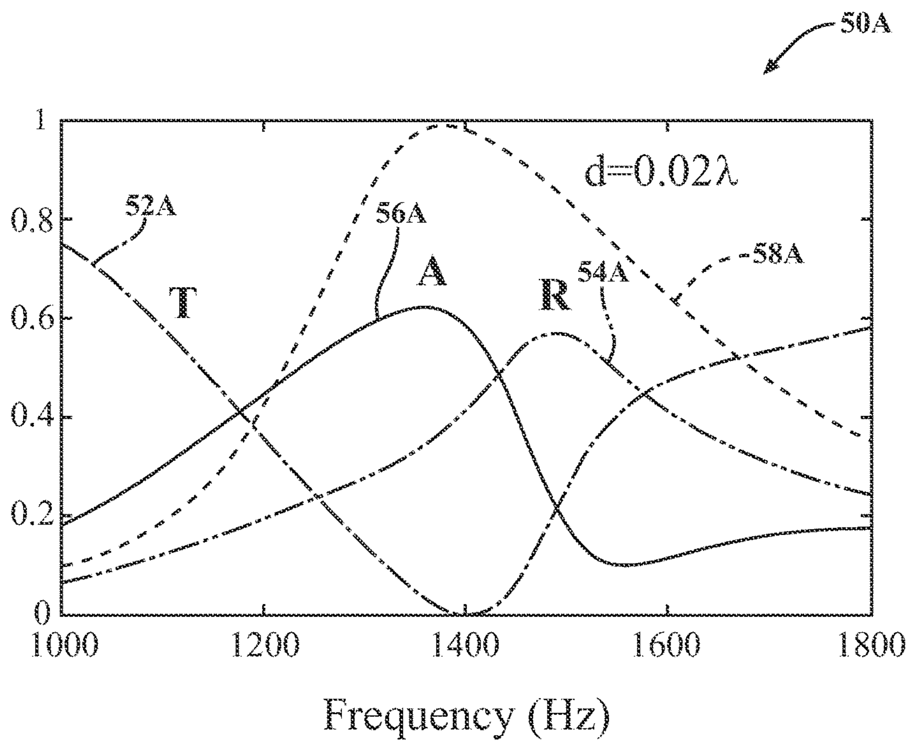
t

FIG. 1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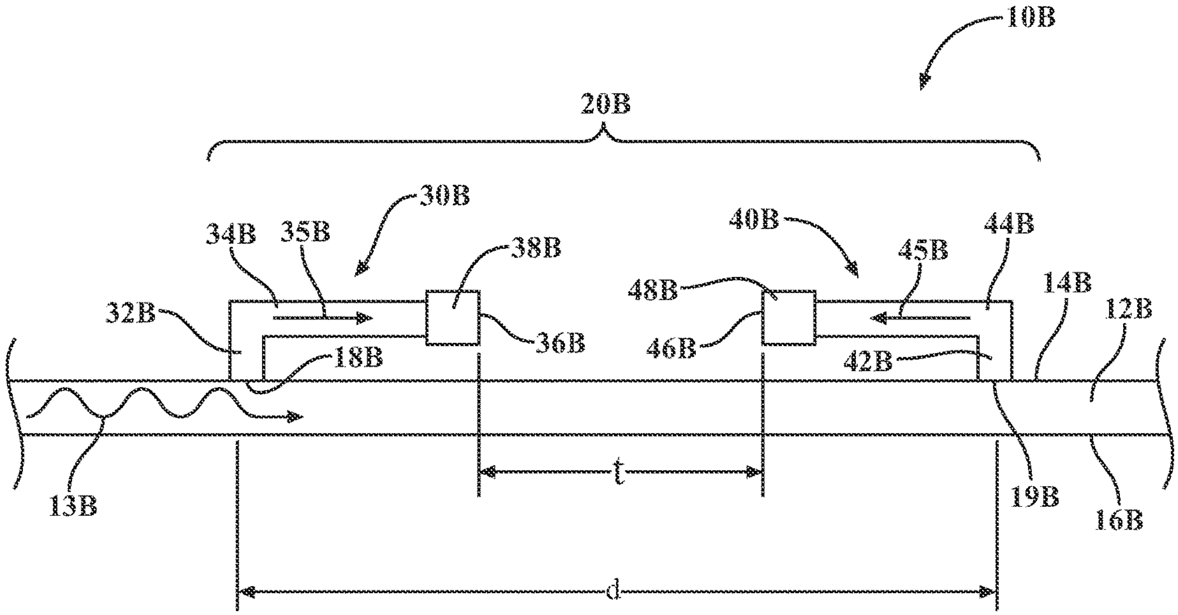
Frequency (Hz)

FIG. 2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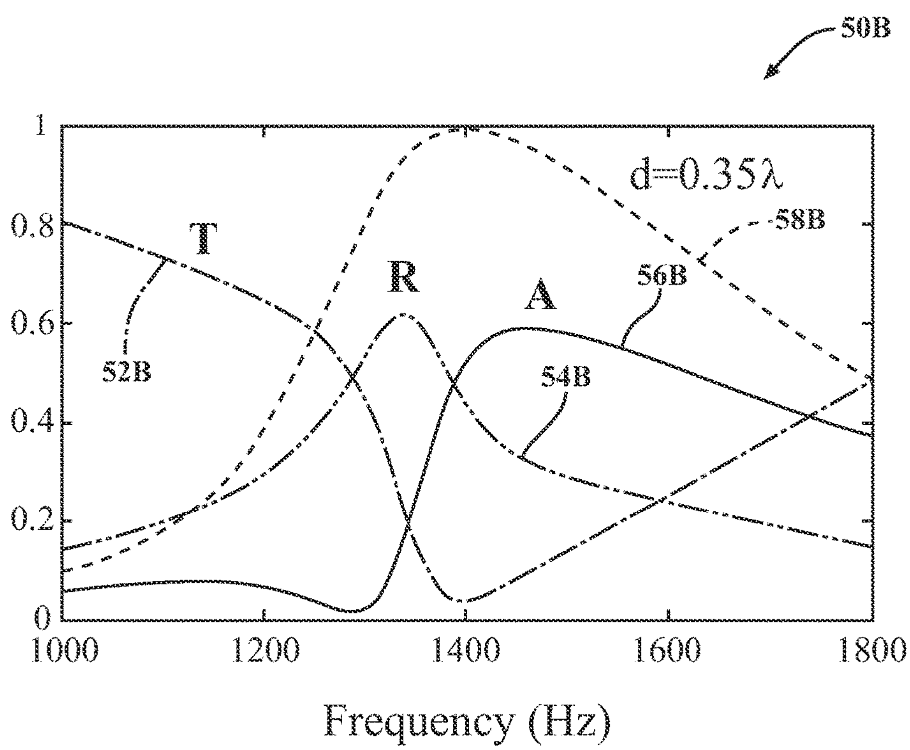
d

FIG. 3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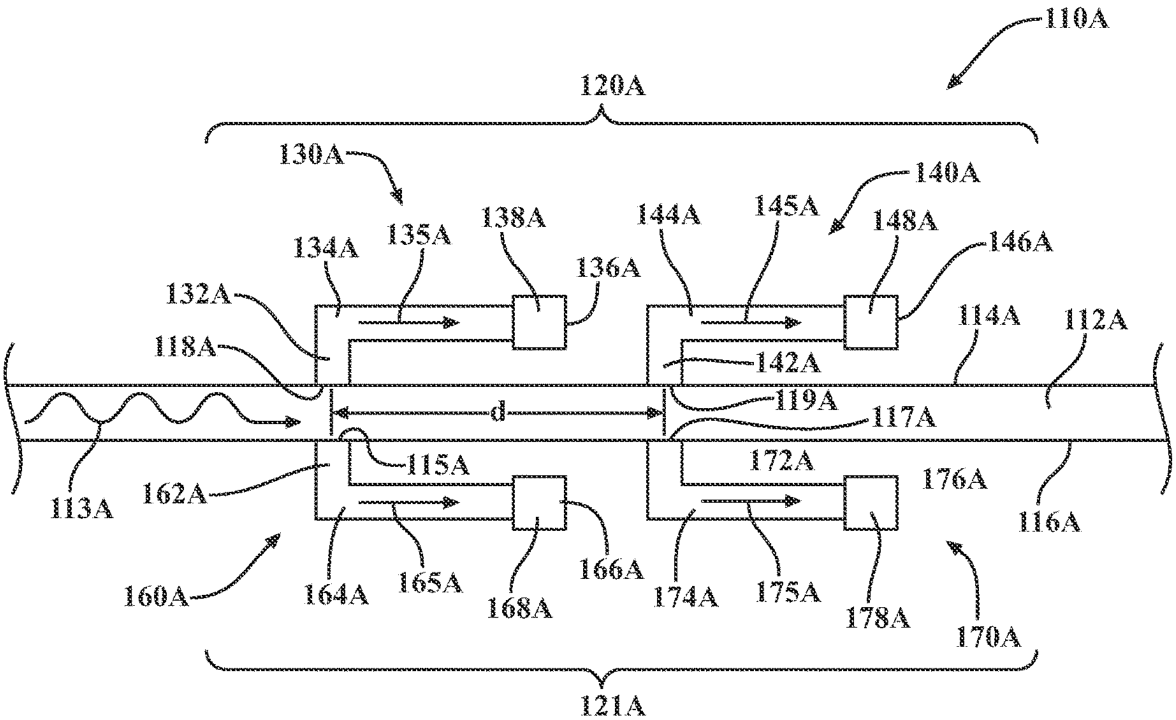
1400
1600
1800
Frequency (Hz)

FIG. 4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112A
d
119A
117A
113A
162A
115A
172A
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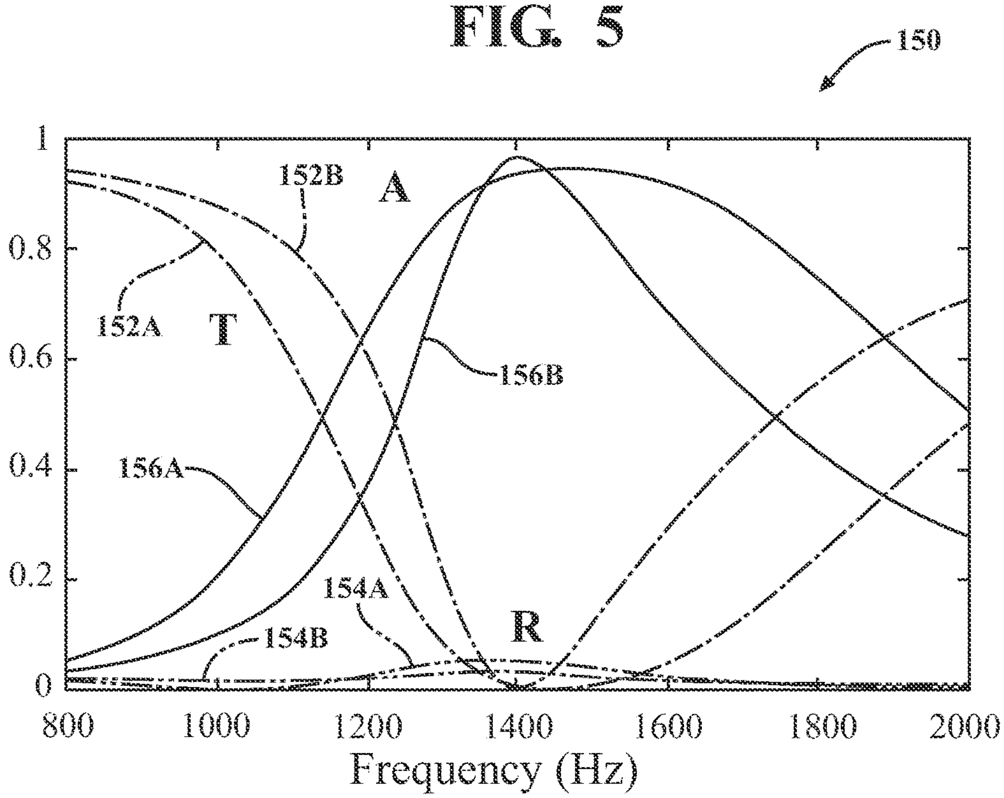
164A
165A
168A
166A
174A
175A
178A
170A
121A

FIG. 5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154B
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0.6
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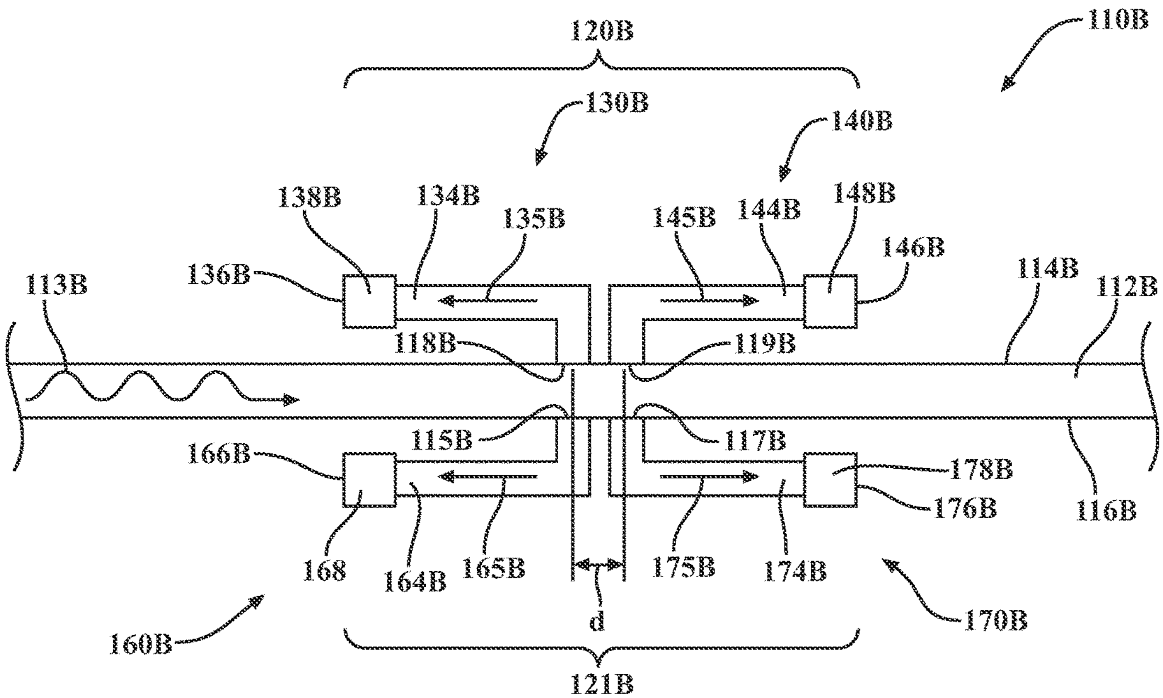
1
800
1000
1200
1400
1600
1800
2000
Frequency (Hz)

FIG. 6



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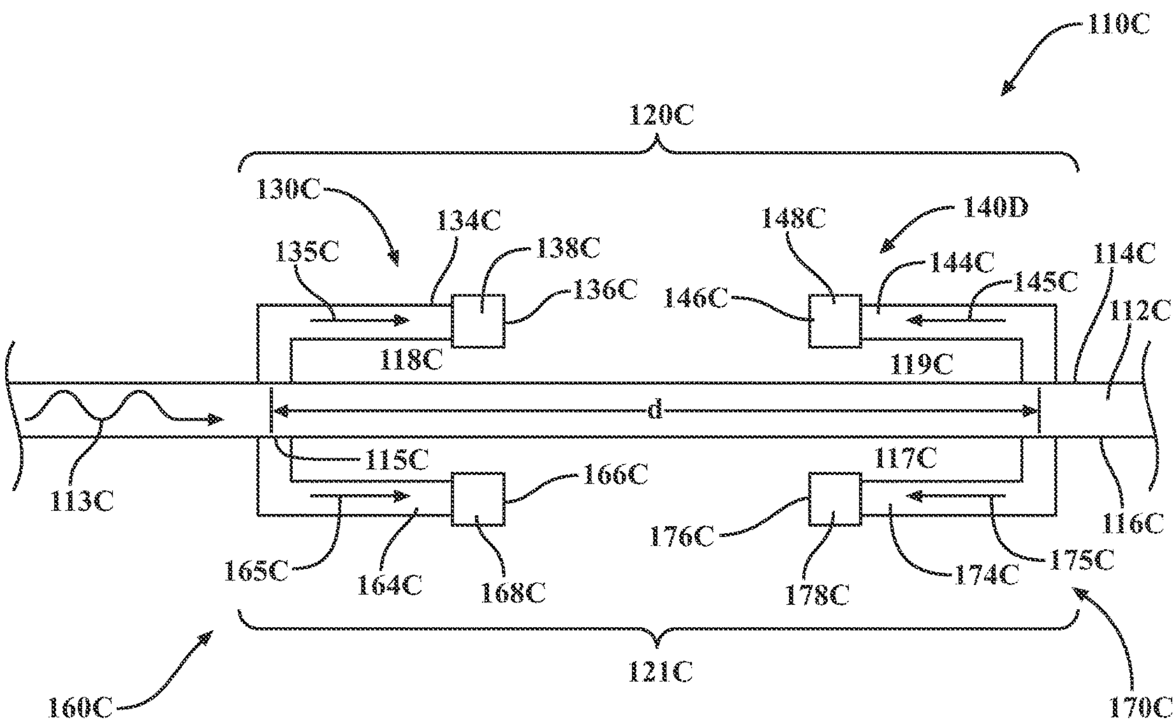
114B
112B
118B
119B
115B
117B
166B
178B
176B
116B
168
164B
165B
175B
174B
d
160B
170B
121B

FIG. 7



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120C
130C
140D
134C
138C
148C
144C
135C
136C
146C
145C
114C
112C
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119C
d
115C
117C
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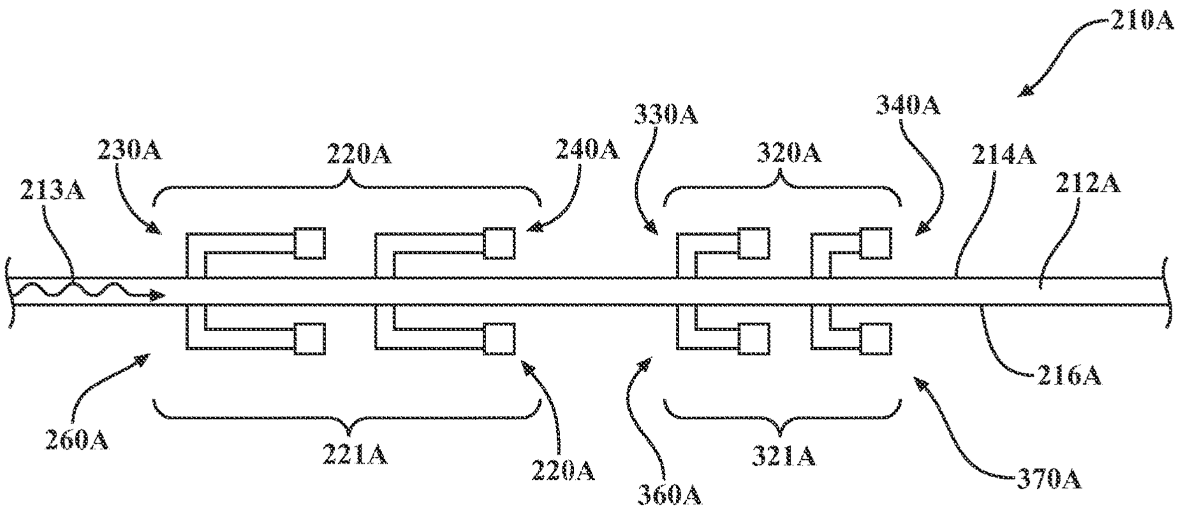
113C
176C
116C
165C
164C
168C
178C
174C
175C
121C
160C
170C

FIG. 8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321A
370A
216A

FIG. 9

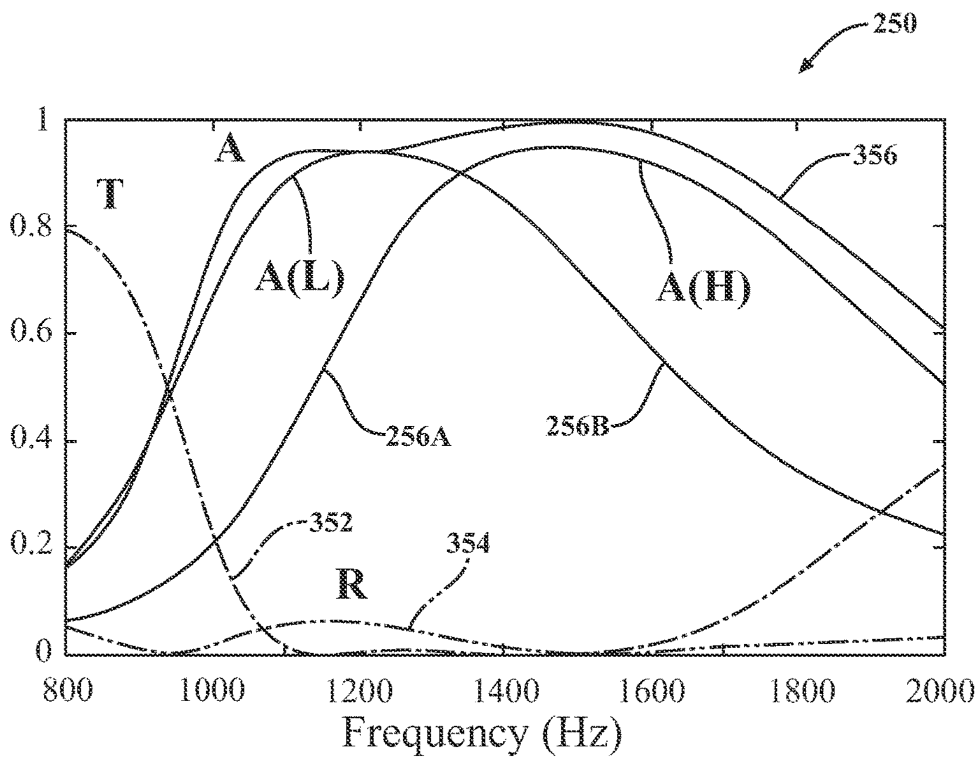
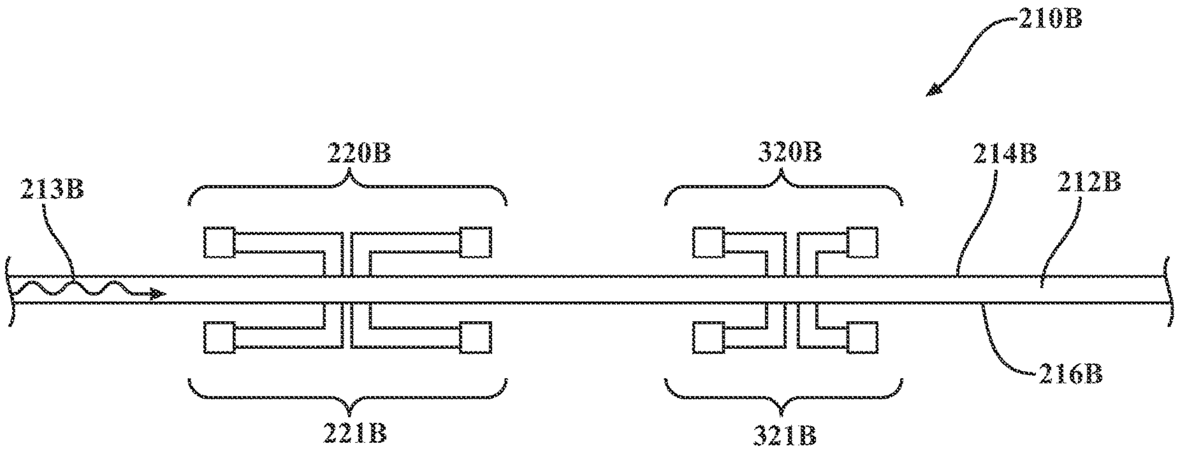

250
1
0.8
0.6
0.4
0.2
0
800
1000
1200
1400
1600
1800
2000
Frequency (Hz)
A
T
A(L)
A(H)
356
256A
256B
352
354
R

FIG. 10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221B
321B

FIG. 11

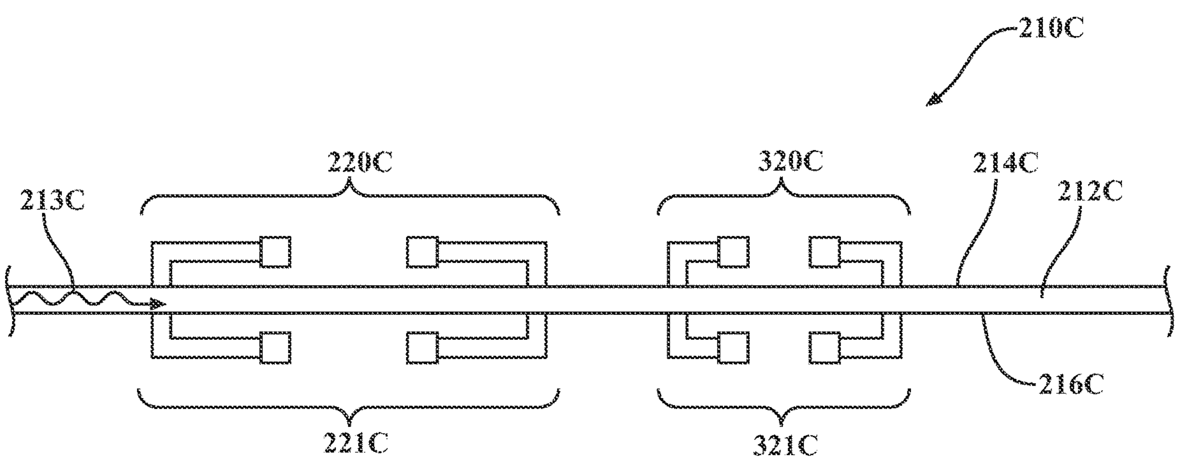
210C
220C
320C
214C
212C
213C
216C
221C
321C

FIG. 12

SYSTEM FOR ABSORBING FLEXURAL WAVES ACTING UPON A STRUCTURE

TECHNICAL FIELD

The subject matter described herein relates, in general, to systems for absorbing flexural waves acting upon a structure.

BACKGROUND

The background description provided is to present the context of the disclosure generally. Work of the inventor, to the extent it may be described in this background section, and aspects of the description that may not otherwise qualify as prior art at the time of filing, are neither expressly nor impliedly admitted as prior art against the present technology.

Some structures, such as beams, are designed to support lateral loads. In doing so, the displacement is predominantly transverse to the centerline, and internal shear forces and bending moments are generated. This dynamic behavior of beams is called flexural motion in the form of flexural waves, sometimes referred to as bending waves. Flexural waves can deform a structure transversely as the flexural waves propagate. These waves are more complicated than compressional or shear waves and depend on the material and geometric properties of the structures they are acting upon. Flexural waves are also dispersive since different frequencies travel at different speeds.

SUMMARY

This section generally summarizes the disclosure and does not comprehensively explain its full scope or all its features.

In one example, a system includes a first resonator connected to a structure at a first location and a second resonator connected to the structure at a second location. The distance between the first and second locations is based on a frequency of a flexural wave acting upon the structure and an orientation of the first resonator and the second resonator with respect to each other.

In another example, a system includes a top set of resonators connected to a top side of a structure. The top set of resonators may be separated from each other at a first distance based on a frequency of a flexural wave acting upon the structure and an orientation of the top set of resonators with respect to each other. The system may also include a bottom set of resonators connected to the bottom side of the structure. The bottom set of resonators are separated from each other at a second distance based on the frequency of the flexural wave acting upon the structure and an orientation of the bottom set of resonators with respect to each other.

In yet another example, the system includes a first set of resonators and a second set of resonators connected to a structure. The first set of resonators are separated from each other at a first distance based on a first frequency of a first flexural wave acting upon the structure and an orientation of the first set of resonators with respect to each other. The second set of resonators are separated from each other at a second distance based on a second frequency of a second flexural wave acting upon the structure and an orientation of the second set of resonators with respect to each other.

Further areas of applicability and various methods of enhancing the disclosed technology will become apparent from the description provided. The description and specific examples in this summary are intended for illustration only and are not intended to limit the scope of the present disclosure.

BRIEF DESCRIPTION OF THE DRAWINGS

The accompanying drawings, which are incorporated in and constitute a part of the specification, illustrate various systems, methods, and other embodiments of the disclosure. It will be appreciated that the illustrated element boundaries (e.g., boxes, groups of boxes, or other shapes) in the figures represent one embodiment of the boundaries. In some embodiments, one element may be designed as multiple elements, or multiple elements may be designed as one element. In some embodiments, an element shown as an internal component of another element may be implemented as an external component and vice versa. Furthermore, elements may not be drawn to scale.

FIG. **1** illustrates a system for absorbing a flexural wave acting upon a structure having resonators that substantially face away from one another.

FIG. **2** illustrates the performance of the system of FIG. **1**.

FIG. **3** illustrates a system for absorbing a flexural wave acting upon a structure with resonators that substantially face one another.

FIG. **4** illustrates the performance of the system of FIG. **3**.

FIG. **5** illustrates a system for absorbing a flexural wave acting upon a structure with resonators substantially facing the same direction connected to both the top and bottom sides of the structure.

FIG. **6** illustrates the performance of the system of FIG. **5**.

FIG. **7** illustrates a system for absorbing a flexural wave acting upon a structure with resonators substantially facing away from each other and connected to both the top and bottom sides of the structure.

FIG. **8** illustrates a system for absorbing a flexural wave acting upon a structure having resonators substantially facing each other and connected to both the top and bottom sides of the structure.

FIG. **9** illustrates a system for absorbing flexural waves acting upon a structure with multiple sets of resonators substantially facing the same direction and connected to both the top and bottom sides of the structure for improved broadband performance.

FIG. **10** illustrates the performance of the system of FIG. **9**.

FIG. **11** illustrates a system for absorbing flexural waves acting upon a structure having multiple sets of resonators substantially facing away from each other and connected to both the top and bottom sides of the structure for improved broadband performance.

FIG. **12** illustrates a system for absorbing flexural waves acting upon a structure with multiple sets of resonators facing each other and connected to both the top and bottom sides of the structure for improved broadband performance.

DETAILED DESCRIPTION

Described herein are examples of systems that can absorb flexural waves acting upon a structure, such as a beam. In one example, a system for absorbing flexural waves acting on a structure includes two resonators connected to the beam at two different locations. Each of the two resonators may be cantilever beam type resonators that include a base connected to the beam and a cantilever member extending from the base. The two resonators may be orientated such that their cantilever members extend from the base substantially toward each other (face toward each other) and/or substantially away from each other (face away from each other).

The distances between where the resonators are connected to the beam can vary based on their orientation with respect to each other and the frequency of the flexural wave to be absorbed. For example, in situations where the resonators face towards each other, the distance where the resonators are connected to the beam will be different than in situations where the resonators face away from each other. Further still, these distances would differ even in situations when the resonators face toward the same direction. Allowing the resonators to face toward or away from each other allows for a more compact and efficient design based on the application and needs regarding the absorption of a flexural wave acting upon the structure.

Referring to FIG. **1**, illustrated is an example of a system **10**A for absorbing a flexural wave. In this example, the system **10**A includes a structure **12**A that may be in the form of a beam. The structure **12**A may include a top side **14**A and a bottom side **16**A. The top side **14**A and/or the bottom side **16**A may be substantially flat but may also be uneven. The structure **12**A can vary significantly from application to application and can take any one of a number of different shapes. Furthermore, the structure **12**A can be made of a number of different materials or combinations of different materials.

In this example, the structure **12**A is shown to have a flexural wave **13**A acting upon the structure **12**A. The flexural wave **13**A can be a single flexural wave or may be multiple flexural waves having the same, similar, or even different frequencies. Flexural waves, sometimes referred to as bending waves, may deform the structure **12**A transversely as they propagate. Flexural waves are more complicated than compressional or shear waves and depend on material properties as well as geometric properties of the structure **12**A.

In the system **10**A, a set **20**A of resonators **30**A and **40**A are connected to the top side **14**A of the structure **12**A. In this example, the resonator **30**A includes a base **32**A connected to the top side **14**A of the structure **12**A at a first location **18**A. The base **32**A may extend upward from the top side **14**A of the structure **12**A to a cantilever member **34**A. The cantilever member **34**A may extend from the base **32**A in a direction **35**A towards a terminal end **36**A, which may include a mass **38**A. In this example, the direction **35**A is substantially parallel to a plane defined by the surface, forming the top side **14**A of the structure **12**A. It should be understood that the direction **35**A may vary from application to application and does not necessarily need to be substantially parallel to a plane defined by the surface of the structure **12**A.

As to the resonator **40**A, the resonator **40**A may be similar to the resonator **30**A in some respects. As such, the resonator **40**A includes a base **42**A connected to the top side **14**A of the structure **12**A at a second location **19**A. The base **42**A may extend upward from the top side **14**A of the structure **12**A to a cantilever member **44**A. The cantilever member **44**A may extend from the base **42**A in a direction **45**A towards a terminal end **46**A, which may include a mass **48**A. In this example, the direction **45**A is substantially parallel to a plane defined by the surface forming the top side **14**A of the structure **12**A. It should be understood that the direction **45**A may vary from application to application and does not necessarily need to be substantially parallel to a surface of the structure **12**A.

It is noted that in this example, the resonators **30**A and **40**A are orientated such that they face away from one another. In other words, the directions **35**A and **45**A in which the cantilever members **34**A and **44**A extend from their bases **32**A and **42**A, respectively, substantially oppose each other and extend away from each other. Additionally, it is noted that one of the resonators **30**A or **40**A may be a lossless resonator, while the other resonator may be a lossy resonator. A lossless resonator is a resonator that can be expressed as a mass-spring system, while a lossy resonator is a resonator that can be expressed as a mass-spring-damper system.

The resonators **30**A and **40**A may each have a resonant frequency that is substantially similar to the frequency of the flexural wave **13**A to be absorbed. The resonant frequency of the resonators **30**A and **40**A is determined by the overall physical dimensions of the resonator and/or the mechanical properties, such as the modulus of the materials that form the resonator. The resonators **30**A and **40**A can be expressed as a mass-spring system, in the case of a lossless resonator, and/or a mass-spring-damper system, in the case of a lossy resonator by considering the first mode of the resonator.

In a situation where one of the resonators **30**A and/or **40**A is a lossless resonator, the following equation is utilized to determine the resonant frequency of the resonator:

$$\omega_0 = \sqrt{\frac{k}{m}}, \tag{1}$$

where ω_0 is the resonant frequency, k is the spring constant, and m is the mass. As such, the overall physical dimensions of the resonator and/or materials forming the resonator will be such that it has a resonant frequency substantially similar to that of the flexural wave **13**A.

In a situation where one of the resonators **30**A and/or **40**A is a lossy resonator, the following equation is utilized to determine the resonant frequency of the resonator as well as determine the overall physical dimensions of the resonator so that the resonator can have a resonant frequency substantially similar to the resonant frequency of the flexural wave **13**A:

$$f_d = f_n\sqrt{1-2\xi^2}, \tag{2}$$

where f_d is the damped resonant frequency, f_n is the undamped resonant frequency and is the damping ratio. The damping ratio can be determined using the following equation:

$$\zeta = \frac{c}{2\sqrt{km}}, \tag{3}$$

where c is the damping coefficient, k is the spring constant, and m is the mass

The distance d between the locations **18**A and **19**A where the bases **32**A and **42**A are connected to the top side **14**A of the structure **12**A, respectively, are based, at least in part, on the frequency of the flexural wave **13**A to be absorbed and the orientation of the resonators **30**A and **40**A with respect to one another. Generally, the distance d between the locations **18**A and **19**A may be expressed as:

$$d = a\lambda, \tag{4}$$

where d between the locations **18**A and **19**A, α is a constant based on observed and or simulation data, and λ is the wavenumber at the frequency of the flexural wave **13**A to be absorbed. The constant α of Equation 4 may be based on the orientation of the resonators **30**A and **40**A with respect to each other and one or more physical characteristics of the structure **12**A. In one example, the constant α is determined based on observation or simulation information, wherein the constant α is the value that maximizes the absorption of the flexural wave **13**A acting upon the structure **12**A.

In situations where the resonators **30**A and **40**A substantially face away from one another, such as in the system **10**A, it has been observed that the distance d between the locations **18**A and **19**A is less than a distance t that represents the distance between the terminal ends **36**A and **46**A. In this example, the distance d can be expressed as 0.02λ. The constant α is 0.02 and was determined based on observation or simulation information that maximized the absorption of the flexural wave **13**A acting upon the structure **12**A.

The performance of the system **10**A is illustrated in the chart **50**A of FIG. **2**. Here, the amount of absorption **58**A of the flexural wave **13**A acting upon the structure **12**A based on the frequency of the flexural wave **13**A is shown. In particular, the system **10**A shows excellent wide bandwidth absorption, peaking at near-total absorption at approximately 1380 Hz. For the sake of comparison, the chart **50**A also illustrates a transmission **52**A, reflection **54**A, absorption **56**A of a system having resonators that substantially face away, but with different distance, d=0.2λ.

As shown in the chart **50**A, the absorption **58**A of the system **10**A is superior to the absorption **56**A of a system with resonators distance, d=0.2λ in both amplitude and bandwidth performance.

FIG. **3** illustrates another example of a system **10**B for absorbing flexural waves. This example of the system **10**B is similar to the system **10**A of FIG. **1**. As such, like reference numerals (replacing "A" with a "B" in the reference numerals) have been utilized to refer to like elements. Unless specifically stated otherwise, the description previously provided regarding these like elements provided previously, is equally applicable to this example.

In this example, the system **10**B includes a set **20**B of resonators **30**B and **40**B connected to the top side **14**B of the structure **12**B. In this example, the resonators **30**B and **40**B are orientated such that they substantially face each other. In other words, the direction **35**B that the cantilever member **34**B extends from the base **32**B and extends towards the other resonator **40**B. Similarly, the direction **45**B that the cantilever member **44**B extends from the base **42**B and extends towards the other resonator **30**B.

Like before, the distance d between the locations **18**B and **19**B where the bases **32**B and **42**B are connected to the top side **14**B of the structure **12**B, respectively, are based, at least in part, on the frequency of the flexural wave **13**B to be absorbed and the orientation of the resonators **30**B and **40**B with respect to one another. Generally, the distance d between the locations **18**B and **19**B can be expressed using Equation 4.

Similarly, the constant α for Equation 4 may be based on the orientation of the resonators **30**B and **40**B with respect to each other and one or more physical characteristics of the structure **12**B. In one example, the constant α is determined based on observation or simulation information, wherein the constant α is the value that maximizes the absorption of the flexural wave **13**B acting upon the structure **12**B.

In situations where the resonators **30**B and **40**B substantially face toward one another, such as in the system **10**B, it has been observed that the distance d between the locations **18**B and **19**B is greater than a distance t that represents the distance between the terminal ends **36**B and **46**B. In this example, the distance d can be expressed as 0.35λ. The constant α is 0.35 and was determined based on observation or simulation information that maximized the absorption of the flexural wave **13**B acting upon the structure **12**B.

The performance of the system **10**B is illustrated in the chart **50**B of FIG. **4**. Here, the amount of absorption **58**B of the flexural wave **13**B acting upon the structure **12**B based on the frequency of the flexural wave **13**B is shown. In particular, the system **10**B shows excellent wide bandwidth absorption, peaking at near-total absorption at approximately 1380 Hz. For the sake of comparison, the chart **50**B also illustrates a transmission **52**B, reflection **54**B, absorption **56**B of a system having resonators that substantially face toward each other with distance, d=0.2λ.

As shown in the chart **50**B, the absorption **58**B of the system **10**B is superior to the absorption **56**B of a system with resonators face toward each other with distance, d=0.2λ in both amplitude and bandwidth performance.

Referring to FIG. **5**, another example of a system **110**A for absorbing flexural waves for broadband absorption is shown. This example of the system **100**A has some similarities with the system **10**A of FIG. **1**. As such, like reference numerals (incremented by 100) have been utilized to refer to like elements. Unless specifically stated otherwise, the description previously provided regarding these like elements provided previously, is equally applicable to this example.

In this example, the system **100**A of FIG. **5** differs from that of the system **10**A of FIG. **1** in at least two ways. First, it is noted that the system **100**A has two sets **120**A and **121**A of resonators, wherein the first set **120**A includes the resonators **130**A and **140**A, while the second set **121**A includes resonators **160**A and **170**A. The first set **120**A of the resonators **130**A and **140**A are connected to the top side **114**A of the structure **112**A, while the second set **121**A of the resonators **160**A and **170**A are connected to the bottom side **116**A of the structure **112**.

Second, it is noted that the resonators **130**A and **140**A substantially face the same direction with respect to each other. In other words, the direction **135**A that the cantilever member **134**A extends from the base **132**A is the same as the direction **145**A that the cantilever member **144**A extends from the base **142**A.

As to the second set **121**A of the resonators **160**A and **170**A, it is noted that these resonators **160**A and **170**A are similar to the resonators **130**A and **140**A, respectively. Notably, the resonator **160**A is connected to the bottom side **116**A of the structure **112**A at a location **115**A, while the resonator **170**A is connected to the bottom side **116**A of the structure **112**A at a location **117**A. Like the resonators **130**A and **140**A, the resonators **160**A and **170**A substantially face the same direction with respect to each other. In other words, the direction **165**A that the cantilever member **164**A extends from the base **162**A is the same as the direction **175**A that the cantilever member **174**A extends from the base **172**A.

Further still, the directions **135**A and **145**A and the directions **165**A and **175**A may all point in the same direction. For example, as noted previously, the resonators **130**A and **140**A face the same direction. The resonators **160**A and **170**A may also face the same direction as the resonators **130**A and **140**A. Visually, resonators **130**A and **140**A are orientated such that they are the mirror image of the resonators **160**A and **170**A. However, it should be understood that the resonators **130**A and **140**A and the resonators **160**A and **170**A may be offset from each other, so they would not be the mirror image of each other.

In this example, one of the resonators **130**A and **140**A forming the first set **120**A is a lossy resonator, while the other may be a lossless resonator. Similarly, one of the resonators **160**A and **170**A forming the second set **121**A may be a lossy resonator, while the other is a lossless resonator. The resonators **130**A, **140**A, **160**A, and/or **170**A may each have a resonant frequency substantially similar to the frequency of the flexural wave **113**A acting upon the structure **112**A.

In the example of the system **100**A, the distance d between the connection locations **118**A and **119**A and the locations **115**A and **117**A is the same. Like before, the distance d may be calculated using Equation 4. The constant α for Equation 4 may be based on the orientation of the first set **120**A of the resonators **130**A and **140**A and the second set **121**A of the resonators **160**A and **170**A with respect to each other and one or more physical characteristics of the structure **112**A. In one example, the constant α is determined based on observation or simulation information, wherein the constant α is the value that maximizes the absorption of the flexural wave **113**A acting upon the structure **112**A.

The performance of the system **110**A is illustrated in the chart **150** of FIG. **6**. Here, the amount of transmission **152**A, reflection **154**A, and absorption **156**A of the flexural wave **113**A acting upon the structure **112**A based on the frequency of the flexural wave **113**A is shown. In particular, the system **110**A shows excellent broadband absorption, peaking at 95% absorption at approximately 1380 Hz. For the sake of comparison, the chart **150** also illustrates a transmission **152**B, reflection **154**B, absorption **156**B of a system having resonators only two resonators, such as the resonators **130**A and **140**A.

As shown in the chart **150**, the absorption **156**A of the system **110**A is nearly as good as the absorption of a hypothetical system only having two resonators but also has significantly improved broadband performance across a much wider range of frequencies.

In the example of the system **110**A, the resonators **130**A, **140**A, **160**A, and **170**A substantially face the same direction. However, it should be understood that other types of systems wherein the resonators face other directions can also be utilized to absorb flexural waves. For example, referring to FIGS. **7** and **8** shown are different examples of systems that can absorb flexural waves at broadband. Like reference numerals (wherein "A" has been changed to "B" or "C") have been utilized to refer to like elements. Unless specifically stated otherwise, the description previously provided regarding these like elements in any of the preceding paragraphs is equally applicable to the examples in FIGS. **7** and **8**.

With particular attention to FIG. **7**, the system **110**B is similar to the system **110**A of FIG. **5** but differs because the resonators **130**B and **140**B face away from each other, similar to the resonators **30**A and **40**A of FIG. **1**. In like manner, the resonators **160**B and **170**B also face away from each other. The distance d between the locations **118**B and **119**B and between the locations **115**B and **117**B may be calculated similarly as previously explained regarding the calculation of the distance d between the locations **18**A and **19**A of the system **10**A of FIG. **1**.

As to FIG. **8**, the system **110**C is similar to the system **110**A of FIG. **5** but differs because the resonators **130**C and **140**C face toward each other, similar to the resonators **30**B and **40**B of FIG. **3**. In like manner, the resonators **160**C and **170**C also face away from each other. The distance d between the locations **118**C and **119**C and between the locations **115**C and **117**C may be calculated similarly as previously explained regarding the calculation of the distance d between the locations **18**B and **19**B of the system **10**B of FIG. **3**.

Variations regarding the examples of the systems described in FIGS. **5**, **7**, and **8** can also be utilized. For example, a system could include a first set of resonators that face in the same direction, such as the first set **120**A of FIG. **5** that are connected to the top side **114**A of the structure **112**A along with a second set of resonators that face towards each other, such as the resonators **160**C and **170**C of the second set **121**C, or that face away from each other, such as the resonators **160**B and **170**B of the second set **121**B. One side of a structure can have resonators orientated in one manner, while the other side of the structure can have resonators orientated differently.

Referring to FIG. **9**, another example of a system **210**A for absorbing flexural waves at improved broadband is shown. This example is somewhat similar to the example of the system **110**A previously described and shown in FIG. **5**. As such, like reference numerals (incremented by 100) have been utilized refer to like elements. Any prior description of these elements is equally applicable to this example unless otherwise noted. Additionally, fewer reference numerals have been utilized in FIG. **9** to improve the overall clarity of the drawings.

The system **210**A includes both a first set **220**A of resonators **230**A and **240**A and a second set **221**A of resonators **260**A and **270**A. The first set **220**A of resonators **230**A and **240**A and a second set **221**A of resonators **260**A and **270**A may be similar to the first set **120**A of resonators **130**A and **140**A and the second set **121**A of resonators **160**A and **170**A of FIG. **5**. As such, the first set **220**A of resonators **230**A and **240**A are attached to the top side **214**A of the structure **212**A, while the second set **221**A of resonators **260**A and **270**A are attached to the bottom side **216**A of the structure **212**A.

The first set **220**A of resonators **230**A and **240**A and the second set **221**A of resonators **260**A and **270**A may be adjusted so that they have a resonant frequency similar to the frequency of a flexural wave **213**A acting upon the structure **212**A. The first set **220**A of resonators **230**A and **240**A the second set **221**A of resonators **260**A and **270**A can be adjusted as described in the paragraphs above to absorb frequencies of flexural waves of a certain frequency range. The adjustment of the first set **220**A of resonators **230**A and **240**A the second set **221**A of resonators **260**A and **270**A can include adjusting the resonant frequencies of these resonators as well as the distances d between where the resonators are connected to the structure **212**A.

However, to improve the broadband performance of the system **210**A, also attached to the structure **212**A are a third set **320**A of resonators **330**A and **340**A and a fourth set **321**A of resonators **360**A and **370**A. The third set **320**A of resonators **330**A and **340**A and the fourth set **321**A of resonators **360**A and **370**A may be somewhat similar to the first set **220**A of resonators **230**A and **240**A and the second set **221**A of resonators **260**A and **270**A. The third set **320**A of resonators **330**A and **340**A are attached to the top side **214**A of the structure **212**A, while the fourth set **321**A of resonators **360**A and **370**A are attached to the bottom side **216**A of the structure **212**A.

However, the third set **320**A of resonators **330**A and **340**A and the fourth set **321**A of resonators **360**A and **370**A are different from the first set **220**A of resonators **230**A and **240**A and the second set **221**A of resonators **260**A and **270**A in that they have been adjusted to have different resonant frequencies for absorbing flexural waves of different frequencies and may be separate from each other at different distances. The third set **320**A of resonators **330**A and **340**A and the fourth set **321**A of resonators **360**A and **370**A may be adjusted to absorb different frequencies of flexural waves acting upon the structure **212**A.

By combining the use of multiple sets of resonators, improved broadband performance can be realized. For example, referring to FIG. **10** illustrated is a chart **250** showing the individual absorption performance **256**A of one set of resonators, such as the first set **220**A of resonators **230**A and **240**A and the second set **221**A of resonators **260**A and **270**A. Also illustrated is the individual absorption performance **256**B of another set of resonators, the third set **320**A of resonators **330**A and **340**A and the fourth set **321**A of resonators **360**A and **370**A. Regarding the absorption performance **256**A, it can be observed that there is good absorption between approximately 1000 Hz and 1400 Hz. Regarding the absorption performance **256**B, it can be observed that there is good absorption performance between approximately 1400 Hz and 1600 Hz.

When implementing multiple sets of resonators, such as shown in the system **210**A, the combined performance of the system can result in improved broadband performance across a much wider range of frequencies. More specifically, because the system **210**A utilizes four different sets **220**A, **221**A, **320**A, and **321**A of resonators, the individual absorption performances **256**A and **256**B can essentially be combined to generate the combined performance **356**. The combined performance **356** illustrates excellent absorption of flexural waves between 1000 Hz and 1600 Hz. Also illustrated is the combined performance of the transmission **352** and the reflection **354** of the system **210**A.

It should be understood that the system **210**A can include multiple sets of resonators and not be limited to just the number of sets of resonators shown in FIG. **9**. As such, the broadband performance of a system can be improved further by adding additional sets of resonators to absorb additional frequencies of flexural waves acting upon a structure. As explained previously, the resonators forming each of the additional sets will need to be adjusted to have different resonant frequencies as well as be adjusted to have different distances between where the resonators are connected to a particular structure, as explained previously in this disclosure.

Further still, it should be understood that the orientation of the resonators with respect to each other can vary from application to application. In the example shown in FIG. **10**, the system **210**A includes resonators **230**A, **240**A, **260**A, **270**A, **330**A, **340**A, **360**A, and **370**A, which all substantially face the same direction. However, it should also be understood that the resonators' orientation can change and can be mixed and matched based on the ultimate application. For example, FIG. **11** illustrates resonator sets **220**B, **221**B, **320**B, and **321**B having resonators that substantially face away from each other, similar to what was shown and described in FIG. **7**. FIG. **12** illustrates resonator sets **220**C, **221**C, **320**C, and **321**C having resonators that substantially face each other, similar to what was shown and described in FIG. **8**.

Additionally, a system could also have sets of resonators wherein some or all of the sets of resonators have different orientations. For example, a system could be devised that includes one set of resonators that face in the same direction, another set of resonators that face in the opposite direction, and another set of resonators that face in the same direction. Again, different types of applications may require different orientations of the resonators in these orientations can vary based on the needs of the application.

The following includes definitions of selected terms employed herein. The definitions include various examples and/or forms of components that fall within the scope of a term and may be used for various implementations. The examples are not intended to be limiting. Both singular and plural forms of terms may be within the definitions.

References to "one embodiment," "an embodiment," "one example," "an example," and so on, indicate that the embodiment(s) or example(s) so described may include a particular feature, structure, characteristic, property, element, or limitation, but that not every embodiment or example necessarily includes that particular feature, structure, characteristic, property, element or limitation. Furthermore, repeated use of the phrase "in one embodiment" does not necessarily refer to the same embodiment, though it may.

The terms "a" and "an," as used herein, are defined as one or more than one. The term "plurality," as used herein, is defined as two or more than two. As used herein, the term "another" is defined as at least a second or more. The terms "including" and/or "having," as used herein, are defined as comprising (i.e., open language). The phrase "at least one of . . . and . . . ." as used herein refers to and encompasses any and all possible combinations of one or more of the associated listed items. As an example, the phrase "at least one of A, B, and C" includes A only, B only, C only, or any combination thereof (e.g., AB, AC, BC, or ABC).

Aspects herein can be embodied in other forms without departing from the spirit or essential attributes thereof. Accordingly, reference should be made to the following claims, rather than to the foregoing specification, as indicating the scope hereof.

What is claimed is:

1. A system comprising:

a first cantilever resonator connected to a flat side of a single beam at a first location;

a second cantilever resonator connected to the flat side of the single beam at a second location; and

wherein a distance between the first location and the second location extends along a length of the flat side of the single beam and is expressed as aλ, wherein λ is a wavenumber at a frequency of a flexural wave to be absorbed and α is 0.02 when the first and second cantilever resonators face away from each other and is 0.2 when the first and second cantilever resonators face toward each other.

2. The system of claim **1**, wherein:

the first cantilever resonator includes a first support base and a first cantilever member that extends in a first direction from the first support base and terminates in a first terminal end; and

the second cantilever resonator includes a second support base and a second cantilever member that extends in a second direction from the second support base and terminates in a second terminal end.

3. The system of claim **2**, wherein an orientation of the first cantilever resonator and the second cantilever resonator with respect to each other is such that the first direction that the first cantilever member extends from the first support base substantially opposes the second direction that the second cantilever member extends from the second support base.

4. The system of claim **3**, wherein the first terminal end and the second terminal end are separated by a distance that is less than the distance between the first location and the second location.

5. The system of claim **3**, wherein the first terminal end and the second terminal end are separated by a distance that is greater than the distance between the first location and the second location.

6. The system of claim **1**, wherein the first cantilever resonator and the second cantilever resonator have resonant frequencies substantially similar to the frequency of the flexural wave acting upon the single beam.

7. The system of claim **6**, wherein the first cantilever resonator is a lossy resonator and the second cantilever resonator is a lossless resonator.

8. A system comprising:

a top set of cantilever resonators connected to a flat top side of a single beam, the top set of cantilever resonators separated from each other by a distance along a length of the flat top side of the single beam; and

a bottom set of cantilever resonators connected to a flat bottom side of the single beam, the bottom set of cantilever resonators separated from each other by the distance along a length of the flat bottom side of the single beam; and

wherein the distance is expressed as aλ, wherein λ is a wavenumber at a frequency of a flexural wave to be absorbed and α is 0.02 when the top set of cantilever resonators face away from each other and is 0.2 when the top set of cantilever resonators face toward each other.

9. The system of claim **8**, wherein resonators forming at least one of the top set of cantilever resonators and the bottom set of cantilever resonators comprise:

a first cantilever resonator having a first support base and a first cantilever member that extends in a first direction from the first support base and terminates in a first terminal end, the first support base being connected to the single beam at a first location;

a second cantilever resonator having a second support base and a second cantilever member that extends in a second direction from the second support base and terminates in a second terminal end, the second support base being connected to the single beam at a second location.

10. The system of claim **9**, wherein an orientation of the first cantilever resonator and the second cantilever resonator with respect to each other is such that the first direction that the first cantilever member from the first support base is such that it substantially opposes the second direction that the second cantilever member from the second support base.

11. The system of claim **9**, wherein the first cantilever resonator and the second cantilever resonator have resonant frequencies substantially similar to the frequency of the flexural wave acting upon the single beam.

12. The system of claim **9**, wherein the first cantilever resonator is a lossy resonator and the second cantilever resonator is a lossless resonator.

13. A system comprising:

a first set of cantilever resonators connected to a single beam, the first set of cantilever resonators separated from each other at a first distance along a length of the single beam, wherein the distance is expressed as aλ, wherein λ is a wavenumber at a first frequency of a first flexural wave to be absorbed and α is 0.02 when the first set of cantilever resonators face away from each other and is 0.2 when the first set of cantilever resonators face toward each other; and

a second set of cantilever resonators connected to the single beam, the second set of cantilever resonators separated from each other at a second distance along the length of the single beam based on a second frequency of a second flexural wave acting upon the single beam and an orientation of the second set of cantilever resonators with respect to each other.

14. The system of claim **13**, wherein cantilever resonators forming at least one of the first set of cantilever resonators and the second set of resonators comprise:

a first cantilever resonator having a first support base and a first cantilever member that extends in a first direction from the first support base and terminates in a first terminal end, the first support base being connected to the single beam at a first location; and

a second cantilever resonator having a second support base and a second cantilever member that extends in a second direction from the second support base and terminates in a second terminal end, the second support base being connected to the single beam at a second location.

15. The system of claim **13**, wherein:

cantilever resonators forming the first set of cantilever resonators have resonant frequencies substantially similar to the first frequency of the first flexural wave acting upon the single beam; and

cantilever resonators forming the second set of cantilever resonators have resonant frequencies substantially similar to the second frequency of the second flexural wave acting upon the single beam.

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